Course Levels: From Clinical Question to Independent Research

Organisation: Neudata · *#ClearDataClearImpact*

Neudata's clinical-research training runs in three levels. Each level builds on the one before, and each department can stop at the level that fits its needs.

	Level 1: Foundations	Level 2: Doing the analysis	Level 3: Advanced topics
In one line	Understand research from question to publication	Fit, check and interpret the common statistical models yourself, in R	Reproducible research, journal-quality output and specialised methods for complex data
Who	A whole department (about 30 health professionals: doctors, nurses, etc.)	An interested subgroup (8 to 12) who want to run their own analyses	Researchers with a specific project or dataset
Prerequisites	None	Level 1 and the free Neudata course *Clinical Data Analysis in R* (or equivalent basics)	Level 2 (or equivalent experience), plus a real question or dataset
Format	8 online sessions of 3 hours, every Saturday for 8 weeks	8 online sessions of 3 hours over about 4 to 6 weeks, with homework	Online workshops of 1 or 2 sessions, or consultancy, on request
Software	None for participants (instructor demos)	R and RStudio on own laptops	R (and specialised packages)
Hands-on?	Concepts, reading and interpreting; analyses demonstrated live on FLUID-ICU	Every session is hands-on coding; the class works together, exercises are individual	Hands-on on the participant's own data
Outcome	An individual research proposal and an individual Results section (other sections optional)	R scripts for an analysis question and a draft Results section	A reproducible analysis, or an analysis plan, for a real project, with Neudata support
Documents	`Course/` (the full Level 1 course)	`level2_hands_on.md`	`level3_advanced.md`

How the levels connect

- **From Level 1 to Level 2.** In Level 1, participants *read* the FLUID-ICU analyses (Table 1, crude vs adjusted ORs, Kaplan-Meier, multiple imputation) that the instructor runs. Level 2 is a statistical-modelling course: participants clean and describe the data, then fit, check and interpret the models that clinical papers use most (linear, logistic, Poisson, negative binomial, zero-inflated and Cox), each session starting with the exploratory data analysis (EDA) suited to that model. It ends with an individual analysis question and a written Results section. The FLUID-ICU dataset and R scripts in `Course/Data/` are shared by both levels.
- **From Level 2 to Level 3.** Level 3 adds the research-craft topics moved out of Level 2 (reproducible projects, version control, Quarto reports, figures that meet journal requirements, and missing-data and sensitivity analyses in depth) and the methods that complex data demand (meta-analysis, prediction models, diagnostic accuracy, advanced survival analysis, longitudinal

and multilevel models, large or clustered datasets). Every Level 3 lecture includes an EDA section and journal-ready figure code for its analysis problem.

Progression at a glance

Skill	Level 1	Level 2	Level 3
Research question, design, ethics	Taught in depth	Assumed	Assumed
Data management and cleaning	Concepts; live demo	Hands-on in R scripts: import, clean, validate, cleaning log	Reproducible projects and version control (Module G); large, linked or clustered data (Module F)
Describing data, Table 1, figures	Read and judge	Table 1; EDA plots suited to each model, from course code	Figures that meet journal requirements (vector vs raster, dimensions, resolution, fonts) and figure code in every module
Regression	Read and interpret outputs	Fit, check, interpret and report linear, logistic, count (Poisson, negative binomial, zero-inflated) and Cox models	Mixed models, competing risks, time-varying covariates, prediction models
Missing data	Mechanisms and ideas	Describe missingness; complete-case vs multiple imputation, touched on	Multiple imputation in practice and sensitivity analyses in depth (Module H)
Survival analysis	Read Kaplan-Meier and HRs	Kaplan-Meier and Cox, including the proportional-hazards check	Advanced: time-varying covariates, competing risks, restricted mean survival time (Module D)
Writing and publishing	The Results section (individual), optional Methods, Introduction, Discussion and Abstract	Results section written from own R output	Quarto reports in which every number comes from code; Methods and Results for complex analyses

Status: Level 1 materials are in `Course/`. Level 2 and Level 3 are outlines. Their materials will be developed after Level 1 has run.